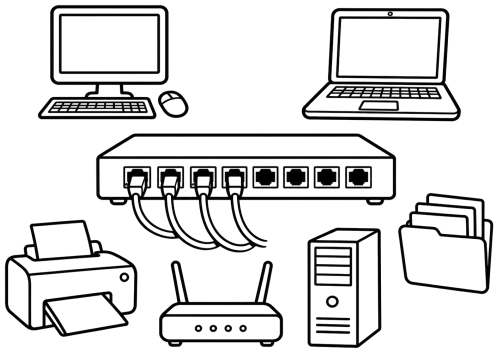
ICT

INFORMATION AND COMMUNICATION TECHNOLOGY

GRADE 9



UNIT 5
Computer Networks

Student Workbook

"The King of ICT"

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STUDENT DETAILS

Name: _____

School: _____

Institution: _____

Contact: _____

BEFORE WE START

What This Unit Is About

Read this page with your teacher

AT THE END OF THIS UNIT YOU WILL BE ABLE TO

- [ ] Explain what a computer network is.
- [ ] List the advantages of computer networking.
- [ ] Name the wired and the wireless media of a network.
- [ ] Name the basic devices of a network and say what each one does.
- [ ] Draw a simple computer network and label its devices.
- [ ] Explain the two basic objectives: communication and sharing resources.
- [ ] Send a message, share a folder and print on a network printer.

THREE RULES FOR THE COMPUTER ROOM

- **Never** plug in or pull out a network cable without your teacher.
- **Always** ask your teacher before you share a folder or a printer.
- **Never** delete or change the files in a friend's shared folder.

HOW TO USE THIS BOOK

Each chapter has three parts. First you **read** the chapter with your teacher. Then you **do** the practical tasks on the network of your computer lab. Last you **write** the exercises in this book.

Colour the small pictures when you finish your work early.



Colour me!

THE WORDS YOU WILL LEARN

Computer network	Resources	Soft copy	Hard copy
Wired	Wireless	Fibre optic	Twisted pair
Infrared	Bluetooth	Wi-Fi	Radio waves
Server	Switch	NIC	RJ45 cable
Router	Internet	Messaging	E-mail
CLI	Licence	Shared folder	Shared printer

"One computer is a tool. Many computers, connected, are a team."

- THE KING OF ICT

CHAPTER 1

What Is a Computer Network?

? CAN I PRINT FROM HERE?

"I need a hard copy of my letter, but the printer is joined to your computer!"

"No problem! Our lab computers are networked. We can print from any computer."

Learn what a **computer network** is, why we use it, and how computers are **connected**.



5.1 What is a computer network?

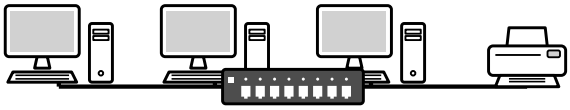
Computers that are **connected together** in a school computer laboratory or in an organisation make a **computer network**. A collection of **two or more computers** connected to **share resources** is a computer network. In a network, the computers share **soft copies** (files) and print **hard copies** (on paper) on one printer.

Stand-alone computers



Only one computer can print.

A computer network



Every computer can print and share files.

5.2 Advantages of computer networking

A computer network has many advantages (textbook Figure 5.1):

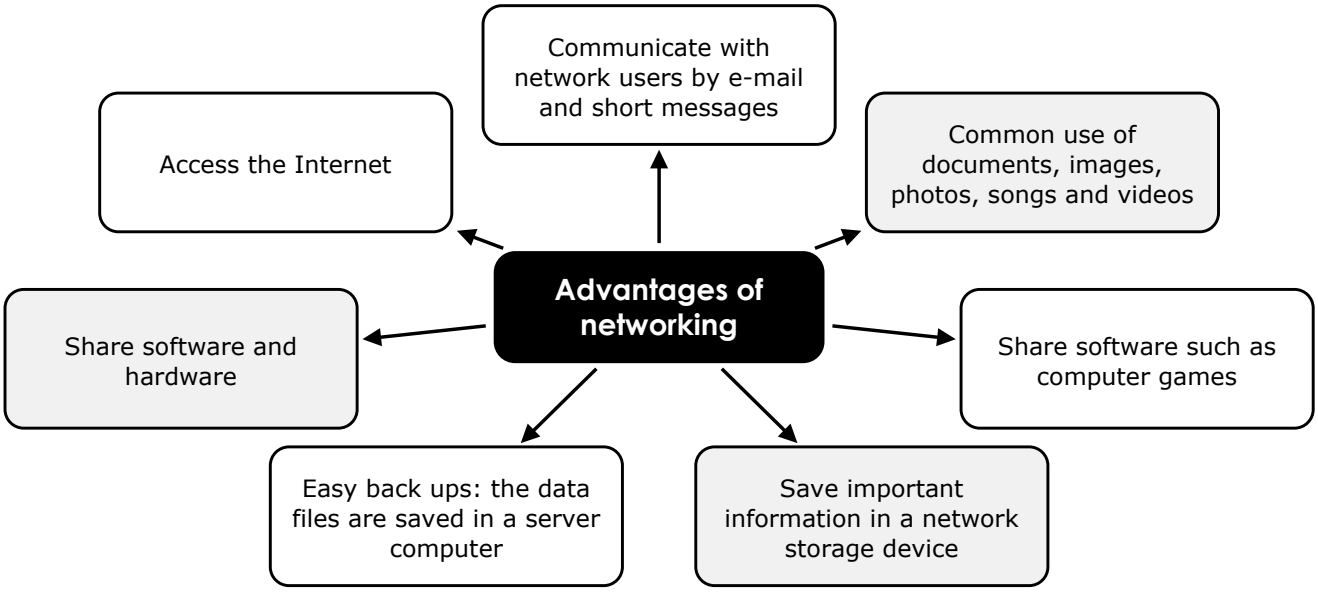
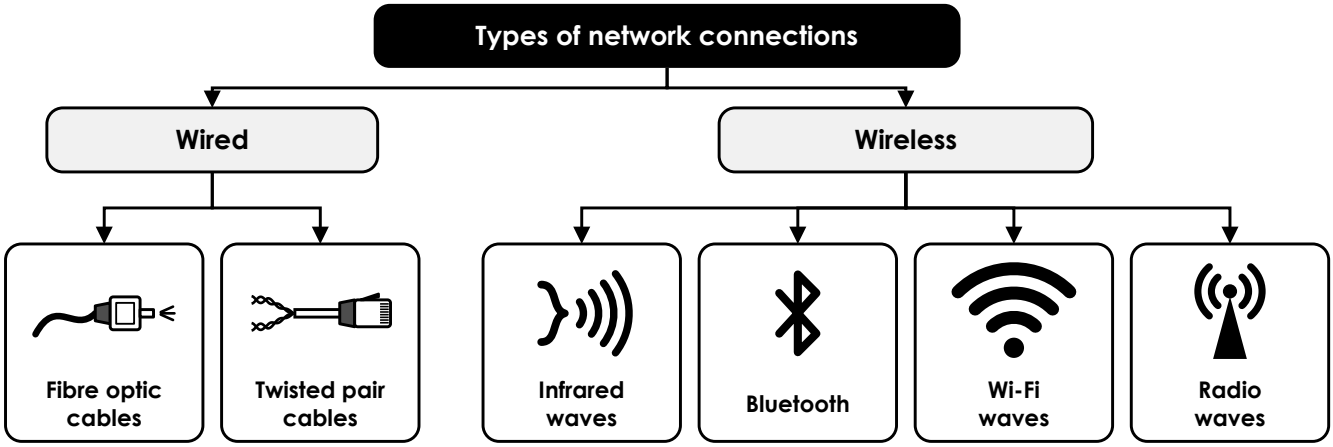


Figure 5.1 - the advantages of using computer networks

5.3 Types of network connections

Computers are connected to a network in two ways: **wired** (with cables) or **wireless** (without cables). Two kinds of cabling are used in wired connections, and there are four examples of wireless media.



Figures 5.2, 5.3 and 5.4 - the types of network connections, the cabling of wired connections and the wireless media

A Wired or wireless?

6 marks

Tick one box for each sentence.

The network connection	Wired	Wireless
Data travels through fibre optic cables.	☐	☐
Two mobile phones share a song with Bluetooth.	☐	☐
The lab computers are joined with twisted pair cables.	☐	☐
A laptop joins the school network with Wi-Fi.	☐	☐
Data is sent with infrared waves.	☐	☐
Data is sent with radio waves.	☐	☐

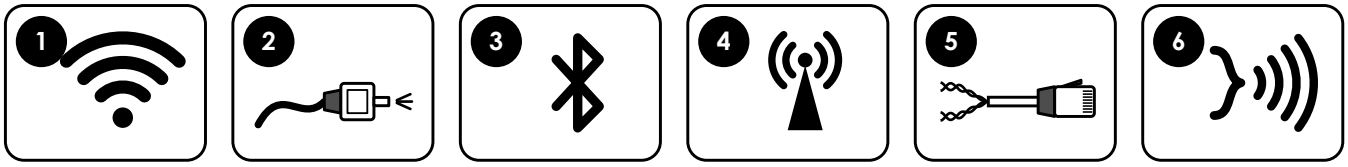
B Name the media

6 marks

Write the name of each medium on the line under it.

WORD BANK

Bluetooth | Fibre optic cable | Infrared waves | Radio waves | Twisted pair cable | Wi-Fi waves



---------- ---------- ---------- ---------- ---------- ----------

C Fill in the blanks

8 marks

Use the words in the box.

WORD BANK

network	two	resources	wireless
twisted pair	stand-alone	radio	hard

1. Computers connected together in a lab make a computer ..... .
2. A computer network has ..... or more computers.
3. The computers of a network are connected to share ..... .
4. The two types of connections are wired and ..... .
5. Fibre optic and ..... cables are used in wired connections.
6. A computer that is not in a network is a ..... computer.
7. Infrared, Bluetooth, Wi-Fi and ..... waves are wireless media.
8. A letter printed on paper is a ..... copy.

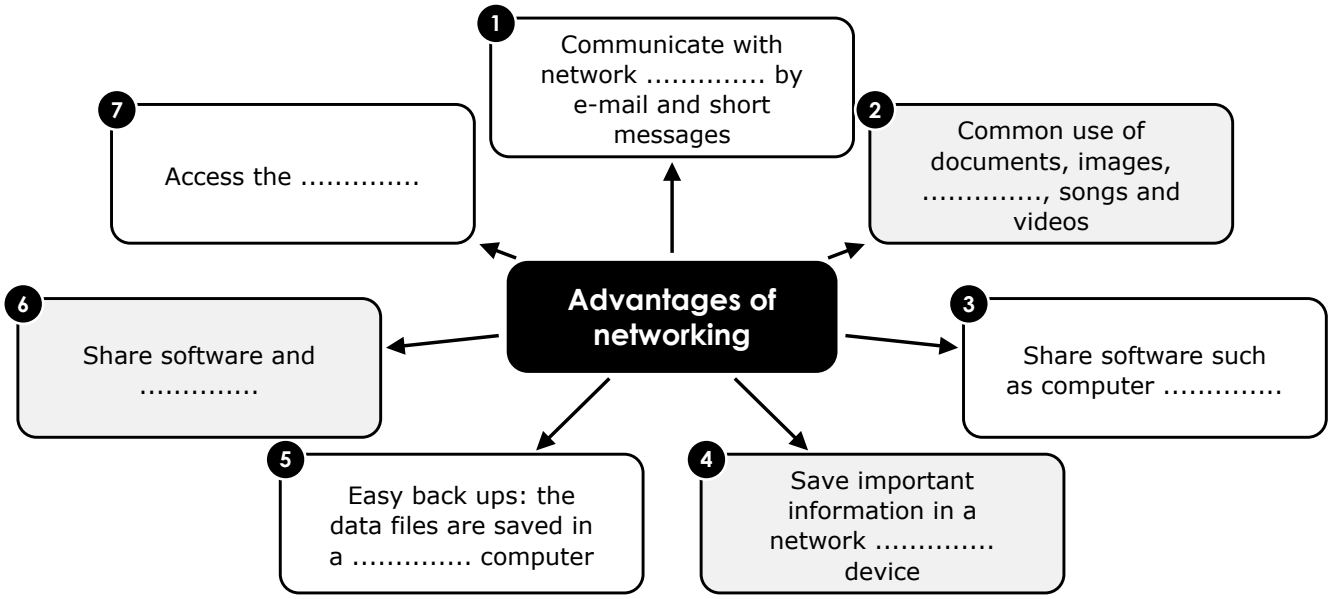
D Complete Figure 5.1

7 marks

Write the missing word in each box. Use the words in the box.

WORD BANK

games hardware Internet photos server storage users



"Connect with good friends. A good network makes you stronger."

- THE KING OF ICT

E True or false

6 marks

Write T or F in the box.

1.		A computer network has only one computer.
2.		In a network, back ups are easy because the data files are saved in a server computer.
3.		Bluetooth is a wired connection.
4.		Fibre optic cables and twisted pair cables are used in wired connections.
5.		The computers of a network can share software such as computer games.
6.		A computer network cannot be used to access the Internet.

F Choose the correct answer

6 marks

Underline your answer.

1. Two or more computers connected to share resources make a computer
   (a) game (b) network (c) screen (d) printer
2. Which one is a wireless medium?
   (a) Fibre optic (b) Twisted pair (c) Wi-Fi waves (d) RJ45 cable
3. Which one is used in wired connections?
   (a) Bluetooth (b) Infrared (c) Radio waves (d) Fibre optic
4. In a network, the data files for back ups are saved in a
   (a) server (b) printer (c) keyboard (d) scanner
5. A copy of a letter printed on paper is a
   (a) soft copy (b) hard copy (c) back up (d) folder
6. How many types of network connections are there?
   (a) one (b) two (c) four (d) seven

G Answer these

4 marks

Write full sentences.

1. Name the two types of network connections. Give one example of each. (2 marks)

2. Your school lab has 20 computers and one printer. Write two advantages of joining them in a network. (2 marks)

CHAPTER 2

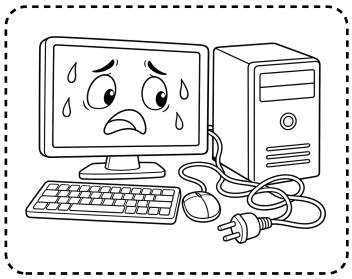
Network Devices

? WHERE DID MY NETWORK GO?

"My computer cannot find the printer or my friend's files today!"

"Look at the back of your computer. The network cable is not plugged in!"

A network needs **devices** to join its computers. Let us meet them.

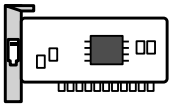
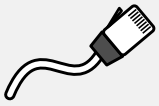


Colour me!

5.4 The basic devices of a network

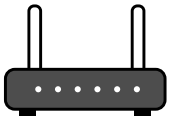
A network needs some **basic devices**: computers with a **Network Interface Card (NIC)**, a **switch** and **connection media** (cables or wireless). A **server** manages the network (textbook Table 5.1).

Table 5.1 - the basic devices of a network

Device	Name	What it does
	**Server**	It manages the **software**, the **hardware**, the **files** and **communication** in the network. Any ordinary computer with the necessary software can be a server. A network may have different servers for different services.
	**Switch**	The **mediator** between two or more computers. It recognises the **destination** computer or device of each message and sends the message there: a print command goes to the **printer** through the switch.
	**Network Interface Card (NIC)**	It connects a computer to a computer network. The network cable is plugged into its port at the back of the computer.
	**RJ45 cables**	Cables with **RJ45 connectors**. They join the computers and other devices to the switch.

5.5 The router

i NOTE - THE ROUTER



The **router** is used to connect one network to **another network** or to the **Internet**. It connects your computer or the network to the Internet easily.

5.6 Two example networks

In Figure 5.5 two computers are joined to a **switch** with **RJ45 cables**. In Figure 5.6 the computers 1 - 4 are connected to the **server** computer **through the switch**.

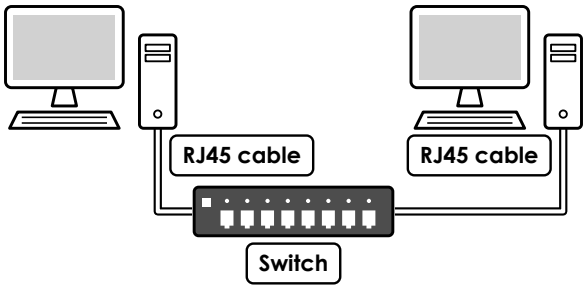


Figure 5.5 - a network involving two computers

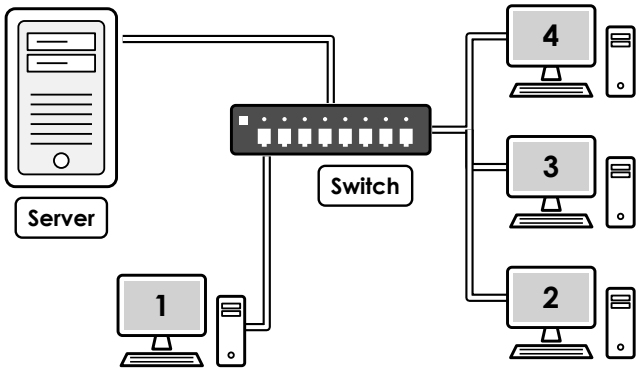


Figure 5.6 - an example of a computer network

DO IT - PRACTICAL 1 (WORKBOOK ACTIVITY 5.1)

Find the network devices of your lab. **Do not pull out any cable.**

1	Find the **Network Interface Card** port at the back of your computer.
2	Look at the cable in that port. Its plug is an **RJ45** connector.
3	Follow the cable with your eyes to the **switch**. Look at its lights.
4	Ask your teacher which computer is the **server**. Where is the **router**?
5	Find the devices that use **wireless** connections (Wi-Fi or Bluetooth).

WRITE WHAT YOU FOUND

Computers in our lab _____ Where our switch is _____

Our server is _____ Wireless devices _____

DO IT - PRACTICAL 2 (WORKBOOK ACTIVITY 5.2)

Build the network of Figure 5.5. **Only your teacher plugs in the cables.**

1	Your teacher brings two computers, a small switch and two RJ45 cables.
2	One cable joins the NIC port of the first computer to the switch.
3	The other cable joins the second computer to the switch.
4	Switch on the switch and both computers. Look at the switch lights.
5	Open **File Explorer** and click **Network**. Is the other computer there?

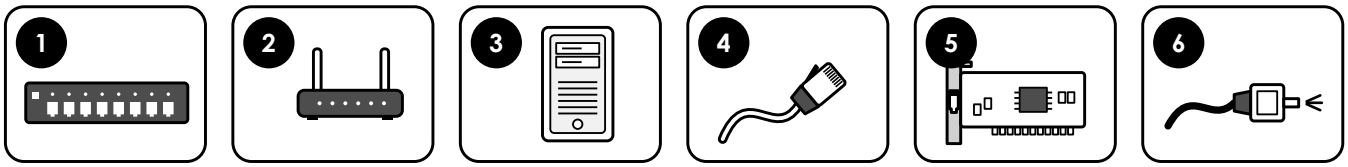
A Name the devices

6 marks

Write the name of each one on the line under it.

WORD BANK

Fibre optic cable	NIC	RJ45 cable	Router	Server	Switch



1. ..... 2. ..... 3. ..... 4. ..... 5. ..... 6. .....

B Who am I?

5 marks

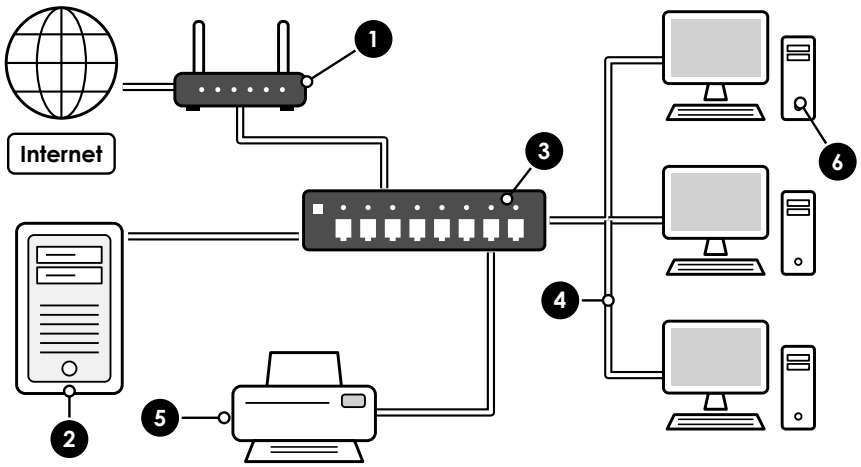
Read each clue. Write the name of the device.

Who am I?	I am the ...
1. I manage the software, hardware, files and communication.	
2. I am the mediator. I find where each message must go.	
3. I am a card. I connect a computer to a computer network.	
4. I connect a network to another network or to the Internet.	
5. I join a computer to the switch. I have RJ45 connectors.	

C Label the network

6 marks

Write the names of the numbered parts 1 - 6.



USE THESE WORDS

Computer	Printer
RJ45 cable	Router
Server	Switch

1. .....
2. .....
3. .....
4. .....
5. .....
6. .....

"Be like a switch: send every message to the right place."

- THE KING OF ICT

D True or false

6 marks

Write T or F in the box.

1.		Any ordinary computer with the necessary software installed can be used as a server.
2.		The switch sends every message to all the computers of the network.
3.		The Network Interface Card connects a computer to a computer network.
4.		A computer network can have only one server.
5.		The router connects one network to another network or to the Internet.
6.		In Figure 5.6 the computers are joined to the server without a switch.

E Draw a network

6 marks

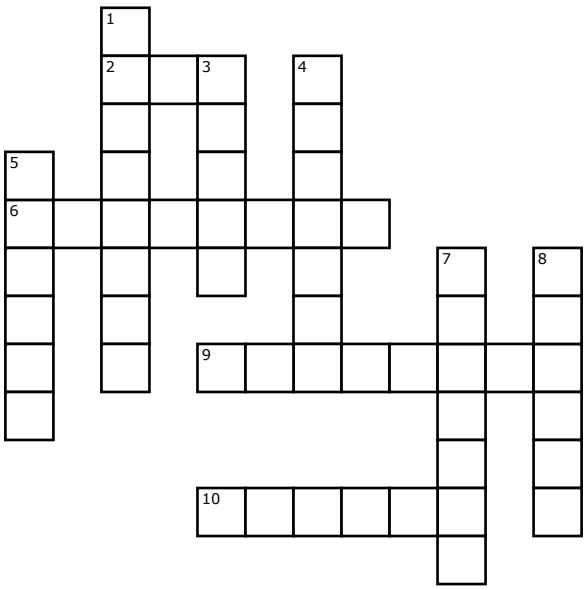
Draw in the box with a pencil.

Draw a network of **two computers** and a **printer** joined by a **switch** (like Figure 5.5). Label the switch and one RJ45 cable.

F Crossword

10 marks

All the words are in Chapters 1 and 2. Write one letter in each square.



ACROSS

2. The short name of the Network Interface Card (3)
6. Bluetooth, Wi-Fi, infrared and radio waves are ... media (8)
9. The switch acts as the ... between two or more computers (8)
10. It connects one network to another network or to the Internet (6)

DOWN

1. A router connects your computer or network to the ... (8)
3. An RJ45 ... joins a computer to the switch (5)
4. ... pair cables are used in wired connections (7)
5. It recognises the destination of a message and sends it there (6)
7. Computers connected together to share resources (7)
8. It manages the software, hardware, files and communication of a network (6)

G Answer these

4 marks

Write full sentences.

1. Nimal gives a print command on Computer 2. How does the command reach the printer? (2 marks)

2. What is the difference between a switch and a router? (2 marks)

CHAPTER 3

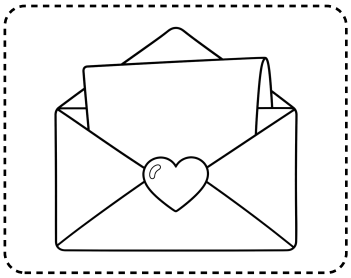
Sharing and Communication

? ONE MESSAGE FOR THE WHOLE CLASS

"Our teacher sent one message to every computer in the lab at the same time!"

"And all of us printed our letters on one printer."

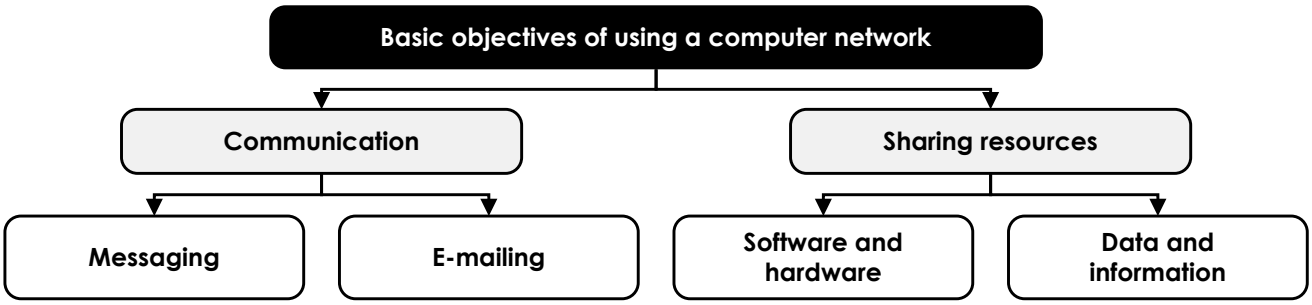
A network has two **basic objectives**: **communication** and **sharing resources**.



Colour me!

5.7 The basic objectives of a network

We use computer networks for two **basic objectives**: **communication** and **sharing resources**.



"A good server serves everyone. A good student helps everyone."

- THE KING OF ICT



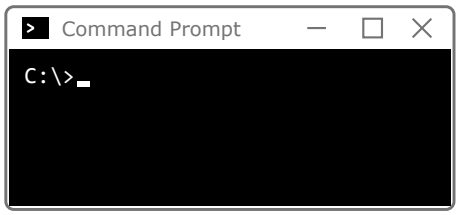
Colour me!

5.8 Communication using a computer network

A computer network helps with easy communication among friends by **messaging** and **e-mailing**. A message can be sent to **many friends** in the network **at one time**. The **Command Line Interface (CLI)** can be used for this.

NOTE - COMMAND LINE INTERFACE (CLI)

The CLI is used to give **typed commands** to the computer, one after another, as **text**. Before the **Graphical User Interface (GUI)** came, the operating system had the command line interface. It can be used to send messages to the network users.

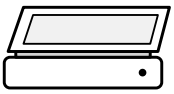


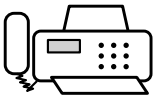
Third party software such as **LAN Messenger**, **Net send GUI** and **POP Messenger** can also be used for network user communications.

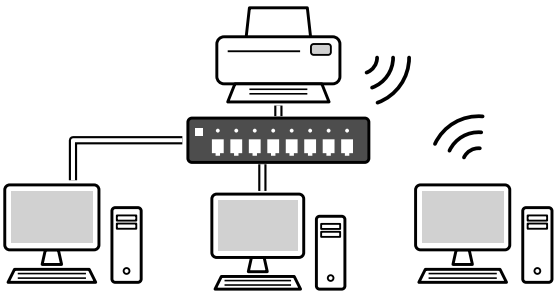
5.9 Sharing resources using a computer network

e.g. 1 - Sharing hardware

When a **printer** is connected to the network with cables or wireless, everyone in the network can use it, when it is set up for sharing. A **scanner** and a **fax machine** can be shared too.


Scanner


Fax machine


A printer shared by the network

ADVANTAGES OF USING PERIPHERALS IN COMMON

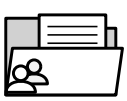
- Several computers can use common hardware.
- It saves money.
- It saves time.

e.g. 2 - SHARING SOFTWARE

Each **stand-alone** computer may need its own software **licence**. This is very expensive. A **multiuser** software licence for a computer network costs much less. This saves a lot of money.

e.g. 3 - DATA AND INFORMATION

Sharing data and information saves **time**, **money** and **storage capacity**, and increases **efficiency** and **productivity**.



When a **folder** is shared in a network, all the files in it can be used by the users of the network. Friends can share images, photos, songs and videos.

DO IT - PRACTICAL 3 (WORKBOOK ACTIVITY 5.3)

Send a message with typed commands. **Do this with your teacher.**

1	Click **Start**, type **cmd** and press **Enter**. The Command Prompt opens.
2	Type the command your teacher gives, e.g. **msg * Hello friends!**
3	Press **Enter**. A box shows your message. Click **OK**.
4	Your teacher sends one message to every computer at one time.
5	Type **exit** and press **Enter** to close the Command Prompt.

DO IT - PRACTICAL 4 (WORKBOOK ACTIVITY 5.4)

Use the messaging software on your computers, e.g. LAN Messenger.

1	Open the messaging software. Find the list of the users of the network.
2	Send **Good morning!** to one friend.
3	Send one message to **many friends** at one time.
4	Read the replies. Which was easier to use: the CLI or the messaging software?

DO IT - PRACTICAL 5 (WORKBOOK ACTIVITY 5.5)

Share a folder in the network. **Ask your teacher before you click Share.**

1	Make a new folder, e.g. **Nimal_Share**, and copy two pictures into it.
2	Right-click the folder and click **Properties**.
3	Open the **Sharing** tab and click **Share...**
4	Choose **Everyone** in the list and click **Add**.
5	Click **Share**, then click **Done**. Your folder is shared.

DO IT - PRACTICAL 6 (WORKBOOK ACTIVITY 5.6)

Open a friend's shared folder. **Never delete or change a friend's files.**

1	Open **File Explorer** and click **Network** on the left.
2	Double-click your friend's computer. Open the shared folder.
3	Copy one picture into your **Pictures** folder and open it.
4	No computers shown? Your teacher turns on network discovery.

"Knowledge is the one thing that grows when you share it."

- THE KING OF ICT

DO IT - PRACTICAL 7 (WORKBOOK ACTIVITY 5.7)

Your teacher shares the printer on the computer it is joined to. Watch.

1	Click **Start** → **Settings** → **Devices** → **Printers & scanners**.
2	Click the printer, then **Manage** → **Printer properties**.
3	Open the **Sharing** tab and tick **Share this printer**.
4	Type a share name, for example **LabPrinter**, and click **OK**.

DO IT - PRACTICAL 8 (WORKBOOK ACTIVITY 5.8)

Print a hard copy on the shared printer from your own computer.

1	Open a short letter that you typed in **Word** (or LibreOffice Writer).
2	Click **File** → **Print**.
3	Choose the shared network printer, for example **LabPrinter**.
4	Not in the list? Your teacher adds it (**Add a printer or scanner**).
5	Print **one** copy. Collect your hard copy from the printer.

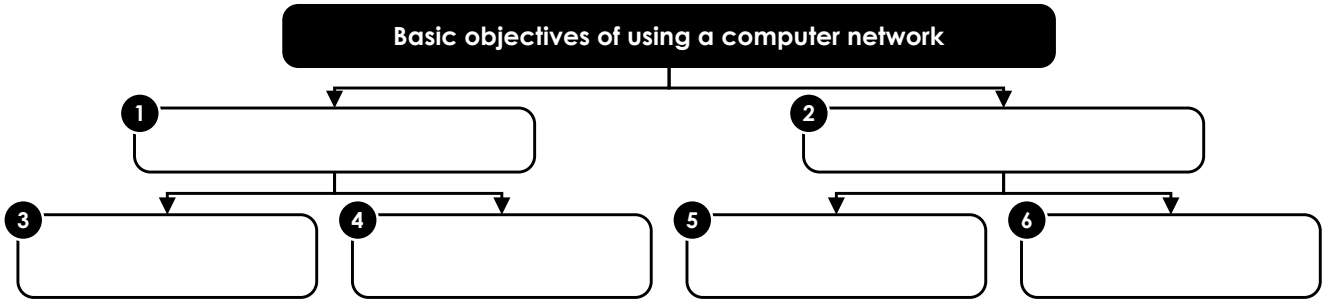
A Complete the tree

6 marks

Write the missing words in the numbered boxes.

WORD BANK

Communication	Data and information	E-mailing
Messaging	Sharing resources	Software and hardware



B Put the steps in order

5 marks

You want to share a folder. Number the steps 1 to 5.

1. ☐ Choose **Everyone** in the list and click **Add**.

2. ☐ Make a new folder and copy your files into it.

3. ☐ Click **Share**, then click **Done**.

4. ☐ Right-click the folder and click **Properties**.

5. ☐ Open the **Sharing** tab and click **Share...**

C What is shared?

8 marks

Is it communication, or sharing hardware, software or data? Tick one box.

In the network ...	Communi-cation	Hardware	Software	Data and information
All the lab computers print on one printer.	☐	☐	☐	☐
The school buys one multiuser licence.	☐	☐	☐	☐
Sports meet photos are in a shared folder.	☐	☐	☐	☐
Sunil sends one message to all his friends.	☐	☐	☐	☐
All the office computers use one scanner.	☐	☐	☐	☐
Students play a game shared on the network.	☐	☐	☐	☐
The teacher e-mails homework to the class.	☐	☐	☐	☐
Concert songs are in a shared folder.	☐	☐	☐	☐

D Fill in the blanks

8 marks

Use the words in the box.

WORD BANK

time	typed	Graphical	licence
money	storage	folder	fax

1. A message can go to many friends at one ..... .
2. The CLI is used to give ..... commands as text.
3. The CLI came before the ..... User Interface.
4. A stand-alone computer may need its own software ..... .
5. Using a printer in common saves ..... and time.
6. Sharing data saves time, money and ..... capacity.
7. All the files in a shared ..... can be used by the users.
8. A scanner and a ..... machine can be shared too.

E Match them

5 marks

Write the correct letter in the box.

A - The word		B - What it is
1. CLI		**A.** All the files in it can be used by the network users.
2. Multiuser licence		**B.** Third party software for messages to network users.
3. Shared folder		**C.** It gives typed commands to the computer as text.
4. Peripherals		**D.** A printer, a scanner and a fax machine.
5. LAN Messenger		**E.** One software licence for the whole network.

F Spot the mistakes

5 marks

Kavindi wrote these notes. Underline the FIVE mistakes and write them correctly.

A computer network has two basic objectives: communication and selling resources. A message can be sent to only one friend at a time. The Command Line Interface gives commands with pictures and icons. When a printer is shared in a network, only one computer can use it. Sharing data and information wastes time, money and storage capacity.

1. ..... Correct: .....
2. ..... Correct: .....
3. ..... Correct: .....
4. ..... Correct: .....
5. ..... Correct: .....

G Word search

10 marks

Find the 10 words. They go across, down and slanting - forwards and backwards.

O	H	T	R	D	U	S	A	K	P	U
C	A	U	I	T	F	I	A	L	S	P
S	R	S	O	F	T	W	A	R	E	M
P	D	L	F	M	F	L	L	L	R	C
R	W	L	I	C	E	N	C	E	L	K
I	A	D	U	I	K	S	B	L	W	F
N	R	E	N	N	A	C	S	I	N	O
T	E	R	C	K	O	H	P	A	K	L
E	N	P	S	B	G	F	D	M	G	D
R	S	H	A	R	I	N	G	E	U	E
C	C	C	K	A	T	A	D	A	R	R

FIND THESE WORDS - TICK EACH ONE

- ☐ SHARING
- ☐ PRINTER
- ☐ SCANNER
- ☐ FOLDER
- ☐ MESSAGE
- ☐ EMAIL
- ☐ SOFTWARE
- ☐ HARDWARE
- ☐ LICENCE
- ☐ DATA

H Answer these

4 marks

Write full sentences.

1. How does sharing software in a network save money? (2 marks)

2. What is a shared folder? Name two things friends can share with it. (2 marks)

REVISION

Unit 5 in One Page

Read this the night before the test

Topic	The key facts
Computer network	Two or more computers connected to share resources, e.g. the computers of the school lab. They share soft copies and print hard copies.
Advantages	Communicate (e-mail, short messages); common use of documents, photos, songs, videos; share software (games) and hardware; network storage; easy back ups on the server; the Internet.
Connections	Wired: fibre optic cables, twisted pair cables. Wireless: infrared waves, Bluetooth, Wi-Fi, radio waves.
Basic devices	Computers with a NIC, a switch and connection media; a server manages the network.
Server	Manages the software, hardware, files and communication. Any ordinary computer with the necessary software can be a server. A network may have different servers.
Switch	The mediator between computers. It recognises the destination of each message (a print command goes to the printer).
NIC and RJ45	The Network Interface Card connects a computer to a network. RJ45 cables join the devices to the switch.
Router	Connects one network to another network or to the Internet.
Objectives	Communication (messaging, e-mailing) and sharing resources (software and hardware, data and information).
CLI	Command Line Interface: typed commands as text, used before the GUI. It can send messages to network users (also LAN Messenger, Net send GUI, POP Messenger).
Sharing	Hardware: printer, scanner, fax machine (saves money and time). Software: a multiuser licence. Data: a shared folder (saves time, money, storage capacity).
In the lab	Share a folder: Properties, Sharing, Share, Everyone, Add, Share. Open it: File Explorer, Network. Print: File, Print, choose the shared printer.

TICK WHEN YOU CAN DO IT

- [ ] I can explain what a computer network is.
- [ ] I can list five advantages of computer networking.
- [ ] I can name the wired and the wireless media.
- [ ] I can name the basic devices of a network and say what each one does.
- [ ] I can draw and label a simple computer network.
- [ ] I can explain communication and sharing resources in a network.
- [ ] I can share a folder and print on a network printer.

WORD POWER

Unit 5 Word List

Twenty-six words you must know

Word	What it means
Computer network	Two or more computers connected together to share resources.
Stand-alone computer	A computer that is not connected to a network.
Resources	What a network shares: hardware, software, data and information.
Soft copy / hard copy	A document on the computer / a document printed on paper.
Wired connection	A connection with cables: fibre optic or twisted pair cables.
Wireless connection	A connection without cables: infrared, Bluetooth, Wi-Fi, radio waves.
Fibre optic cable	A cable used in wired connections (the textbook writes "fiber").
Twisted pair cable	A cable used in wired connections; the network cable of a lab.
Server	Manages the software, hardware, files and communication in a network.
Switch	The mediator between computers: it sends each message to its destination.
Mediator	A go-between. The switch is the mediator between two or more computers.
Destination	The computer or device that a message is sent to.
NIC	Network Interface Card: it connects a computer to a computer network.
RJ45 cable	A network cable with RJ45 connectors; it joins a device to the switch.
Router	Connects one network to another network or to the Internet.
Internet	The network that a router connects your computer or network to.
Communication	Messaging and e-mailing between the users of a network.
Messaging	Sending messages to one friend, or to many friends at one time.
CLI	Command Line Interface: it gives typed commands to the computer as text.
GUI	Graphical User Interface: it came after the CLI.
Peripheral	A device used with a computer, such as a printer, scanner or fax machine.
Software licence	Permission to use software. A multiuser licence is for a whole network.
Shared folder	A folder whose files can be used by all the users of a network.
Shared printer	A printer that every computer of the network can use.
Back up	A safe copy of data files, for example on the server.
Productivity	How much work is done. Sharing data and information increases it.

"Every network has its own language. Learn its words and you can join in."

- THE KING OF ICT

END OF UNIT

Unit 5 Test

Time: 45 minutes Total: 40 marks

Name _____ **Class** _____ **Marks** ____

1 Choose the correct answer

8 marks

Underline your answer.

1. Which one is NOT a basic device of a computer network?
   (a) Switch (b) NIC (c) Server (d) Keyboard
2. Which one is a wireless medium?
   (a) Fibre optic (b) Bluetooth (c) Twisted pair (d) RJ45 cable
3. Which device connects a computer to a computer network?
   (a) NIC (b) Scanner (c) Printer (d) Monitor
4. A print command goes to the printer through the
   (a) switch (b) router (c) scanner (d) fax machine
5. Which device connects a network to the Internet?
   (a) Switch (b) Router (c) Printer (d) Scanner
6. Communication and ... resources are the two basic objectives of a network.
   (a) buying (b) sharing (c) printing (d) selling
7. The interface that is used to give typed commands as text is the
   (a) GUI (b) CLI (c) NIC (d) Wi-Fi
8. A - Any ordinary computer with the necessary software can be a server. B - A multiuser licence costs more than a licence for every stand-alone computer. C - All the files in a shared folder can be used by the network users. Which are correct?
   (a) A and B only (b) A and C only (c) B and C only (d) A, B and C

2 Fill in the blanks

8 marks

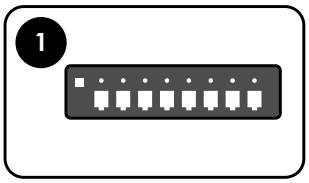
One word in each blank.

1. Fibre optic cables and ..... pair cables are wired media.
2. Infrared, Bluetooth, Wi-Fi and ..... waves are wireless media.
3. The ..... manages the software, hardware and files.
4. RJ45 ..... join the computers to the switch.
5. Messaging and ..... help network users to communicate.
6. A printer, a scanner and a ..... machine can be shared.
7. Sharing data saves time, money and ..... capacity.
8. Before the GUI, computers had the Command ..... Interface.

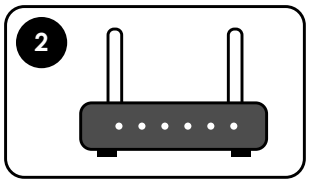
3 Look at the pictures

8 marks

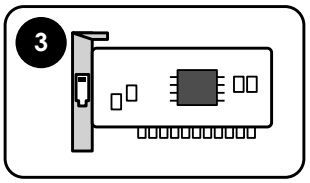
Write the name of each one, and what it does or what type it is.



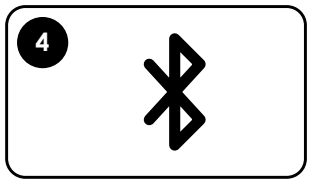
Name: .....



Name: .....



Name: .....



Name: .....

4 Answer these

16 marks

Write in the space given.

1. What is a computer network? (2 marks)

2. Write three advantages of computer networking. (3 marks)

3. Name the two types of network connections. Give two examples of the media of each type. (3 marks)

4. Draw a network of three computers connected to a server through a switch. Label the server and the switch. (3 marks)

5. What are the two basic objectives of a computer network? (2 marks)

6. Write three advantages of using a printer in common in a network. (3 marks)